ENVS-193DS Final

Gabriela Davidson

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Link to Git Hub

GitHub Repository

Set Up

```
library(tidyverse)   # data wrangling and ggplot
library(here)        # file paths
library(janitor)     # data cleaning
library(DHARMa)      # diagnostic checks for models
library(MuMIn)       # AIC model selection
library(ggeffects)   # model predictions
library(flextable)   # making tables
library(rstatix)     # effect size calculations
```

```
nest_data <- read_csv(here("data", "nest_data_final.csv"))
my_data <- read_csv(here("data", "my_data.csv"))
```

Problem 1. Research writing

1a. Transparent statistical methods

Part 1:

In part 1, they used a logistic regression, since the response variable (probability of American Avocet micro habitat use) is binary and distance to water edge is a continuous predictor. #### Part 2: In part 2, they used a chi-square test of independence, since they are testing the relationship between two categorical variables (bird species and habitat type).

1b. Suggestions for rewriting

Part 1: Distance to water edge was a significant predictor of American Avocet microhabitat use, meaning that where a bird was found was related to how far it was from the water's edge.

Part 2: The three bird species, American Avocet, Forster's Tern, and Black-necked Stilt, were not evenly distributed across habitat types (managed pond, non-tidal marsh, salt pond, tidal marsh, and tidal mudflat), suggesting that each species shows preferences for particular habitats.

1c. Further analysis

Presence of wet bare ground (categorical: present or absent, assessed visually during point count surveys) could influence the probability of American Avocet microhabitat use because, according to Schacter et al. (2023), Avocets fed primarily on wet bare ground in 65% of foraging observations, suggesting they strongly select for this substrate type when choosing where to forage. Sites lacking wet bare ground would therefore be expected to have lower probability of Avocet microhabitat use.

Problem 2. Data analysis

2a. Clean and Wrangle

```r
nests_clean <- nest_data |>
  # select only the 3 columns of interest
  select(case_control, height_cm, ant_species) |>
  # filter to only the 3 ant species of interest
  filter(ant_species %in% c("AB", "RRB", "BBR")) |>
  # create column with full scientific names
  mutate(ant_species_full = case_when(
    ant_species == "AB"  ~ "Crematogaster sjostedti",
    ant_species == "RRB" ~ "Crematogaster mimosae",
    ant_species == "BBR" ~ "Crematogaster nigriceps"
  )) |>
  # reorder factor levels: C. sjostedti, C. mimosae, C. nigriceps
  mutate(ant_species_full = fct_relevel(ant_species_full,
                                        "Crematogaster sjostedti",
                                        "Crematogaster mimosae",
                                        "Crematogaster nigriceps"))
```

```r
# display structure of nests_clean
str(nests_clean)
```

```
tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ case_control    : num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ height_cm       : num [1:270] 392 310 513 154 324 ...
 $ ant_species     : chr [1:270] "RRB" "RRB" "RRB" "RRB" ...
 $ ant_species_full: Factor w/ 3 levels "Crematogaster sjostedti",..: 2 2 2 2 1 2 3 2 1 3 ..
```

```r
# display 10 random rows
slice_sample(nests_clean, n = 10)
```

```
# A tibble: 10 x 4
   case_control height_cm ant_species ant_species_full
          <dbl>     <dbl> <chr>       <fct>
 1            0     116.  RRB         Crematogaster mimosae
 2            0     212.  RRB         Crematogaster mimosae
 3            0      96.5 BBR         Crematogaster nigriceps
 4            0     522   RRB         Crematogaster mimosae
 5            0     165   RRB         Crematogaster mimosae
 6            1     191   RRB         Crematogaster mimosae
 7            0     276.  RRB         Crematogaster mimosae
 8            0     166.  RRB         Crematogaster mimosae
 9            0     144.  RRB         Crematogaster mimosae
10            1     234.  BBR         Crematogaster nigriceps
```

2b. Response variable

In this dataset, a 1 indicates that the tree was a "used" tree, meaning a bird nest of any species was found in that tree. A 0 indicates that the tree was an "available" tree, meaning it was one of the four nearest neighbors to a nest tree but did not itself contain a nest.

2c. Purpose of study

Tree height is a continuous variable measured in centimeters. As tree height increases, the probability of nest occurrence would increase because taller trees may offer greater structural complexity and concealment from predators, making them more suitable nest sites; the authors found that for each 1 m increase in height, the odds that birds nested in a tree increased by 20%. This information was found in the Methods section (paragraph 2) and the Results section (paragraph 1).

Acacia ant species is a categorical variable with three levels: Crematogaster sjostedti, Crematogaster mimosae, and Crematogaster nigriceps. As ant species identity shifts from less aggressive defenders (C. sjostedti) to more aggressive defenders (C. mimosae and C. nigriceps), the probability of nest occurrence would increase because more aggressive ant species vigorously defend host trees against mammalian herbivores and predators, which may also protect bird nests from disturbance and predation. This information was found in the Introduction (paragraphs 2–3) and the Discussion (paragraphs 1–2).

2d. Table of Models

```
# create data frame with model information for table
model_table <- data.frame(
  # model numbers as character strings
  "Model Number" = c("1", "2"),
  # response variable is the same for both models
  "Response Variable" = c("Probability of nest occurrence",
                          "Probability of nest occurrence"),
  # both models use the same predictors
  "Predictor Variables" = c("Tree height (cm), ant species",
                            "Tree height (cm), ant species"),
  # model 1 has interaction term, model 2 does not
  "Interaction Term" = c("Yes (height × ant species)",
                         "No")
)

# convert data frame to flextable for formatted output
flextable(model_table) |>
  # rename column headers to remove periods added by data.frame()
  set_header_labels(
    Model.Number = "Model Number",
    Response.Variable = "Response Variable",
    Predictor.Variables = "Predictor Variables",
    Interaction.Term = "Interaction Term"
  ) |>
  # center all text in all cells
  align(align = "center", part = "all") |>
  # left align the first column (model number) only
  align(j = 1, align = "left", part = "all") |>
  # bold the header row
  bold(part = "header") |>
  # add gray background to header row
```

```r
  bg(part = "header", bg = "#D3D3D3") |>
  # add horizontal borders between rows in the body
  border_inner_h(part = "body") |>
  # set table to full width and autofit columns, reduce cell padding for pdf
  set_table_properties(width = 1, layout = "autofit",
                       opts_pdf = list(tabcolsep = 3)) |>
  # set font size to 9 for all parts of the table
  fontsize(size = 9, part = "all") |>
  # italicize the predictor variables column (column 3) in the body
  italic(j = 3, part = "body") |>
  # prevent table from splitting across pages in pdf output
  paginate(init = TRUE, hdr_ftr = TRUE)
```

Model Number	Response Variable	Predictor Variables	Interaction Term
1	Probability of nest occurrence	*Tree height (cm), ant species*	Yes (height × ant species)
2	Probability of nest occurrence	*Tree height (cm), ant species*	No

Table 1. Overview of the two candidate models predicting probability of bird nest occurrence. Both models include tree height (cm) and acacia ant species as predictors. Model 1 includes an interaction term allowing the effect of tree height to vary by ant species, while Model 2 assumes the effect of tree height is consistent across all ant species.

2e. Run models

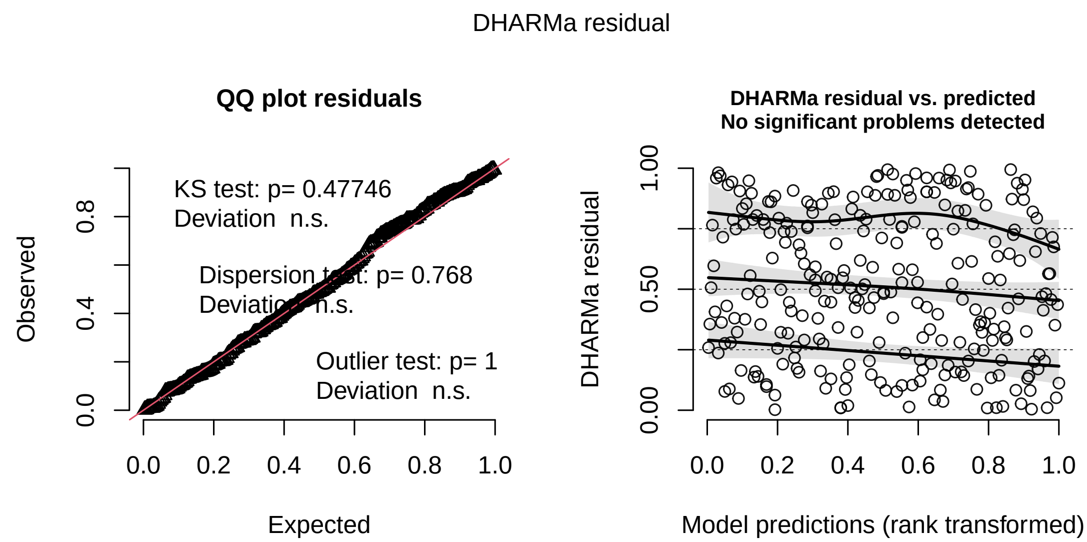
```r
# model 1: tree height + ant species, no interaction
model1 <- glm(case_control ~ height_cm + ant_species_full,
              data = nests_clean,
              family = binomial)

# model 2: tree height * ant species, with interaction
model2 <- glm(case_control ~ height_cm * ant_species_full,
              data = nests_clean,
              family = binomial)
```

2f. Select the best model

The best model as determined by Akaike's Information Criterion (AIC) was the model with an interaction term between tree height and acacia ant species, predicting the probability of bird nest occurrence (AICc = 238.12). This model performed better than the additive model with no interaction term (AICc = 258.89, ΔAICc = 20.77), indicating that the effect of tree height on the probability of nest occurrence differs depending on which acacia ant species occupies the tree. A ΔAICc greater than 2 indicates meaningful improvement in model fit, suggesting the interaction term is an important component of the model.

2g. Check diagnostics

```
# simulate residuals from best model (model 2, with interaction)
simulation_output <- simulateResiduals(fittedModel = model2,
                                       plot = TRUE)
```

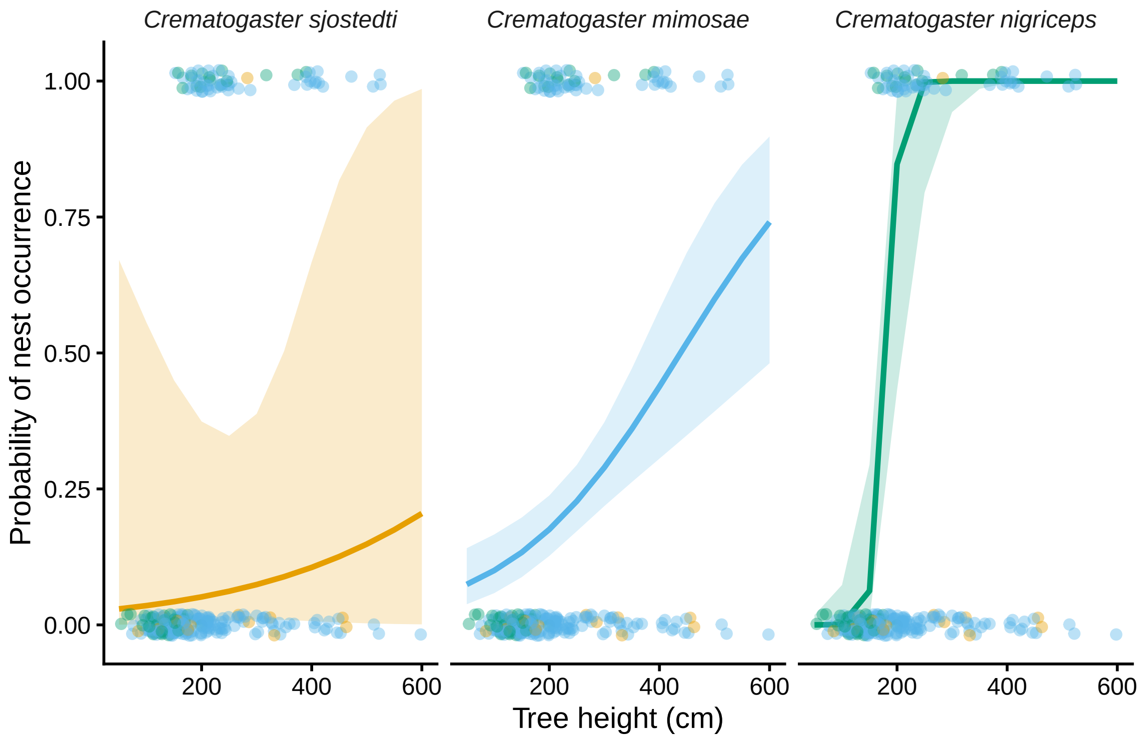


The QQ plot shows simulated residuals closely following the expected distribution, and the residual vs. predicted plot shows no significant problems detected (KS test: p = 0.48, dispersion test: p = 0.77, outlier test: p = 1). The model assumptions are met and the model is appropriate for inference.

2h. Visualize the model predictions

```
# generate predictions from model 2 for height across all 3 ant species
predictions <- ggpredict(model2,
                         terms = c("height_cm", "ant_species_full"))
```

```r
# plot predictions with underlying data
ggplot(predictions, aes(x = x, y = predicted, color = group, fill = group)) +
  # 95% confidence interval ribbon
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high),
              alpha = 0.2,
              color = NA) +
  # model prediction line
  geom_line(linewidth = 1) +
  # underlying data points, jittered slightly so 0s and 1s are visible
  geom_jitter(data = nests_clean,
              aes(x = height_cm,
                  y = case_control,
                  color = ant_species_full),
              height = 0.02,
              alpha = 0.4,
              inherit.aes = FALSE) +
  # separate panel per ant species
  facet_wrap(~group) +
  # custom colorblind-friendly colors
  scale_color_manual(values = c("Crematogaster sjostedti" = "#E69F00",
                                "Crematogaster mimosae"   = "#56B4E9",
                                "Crematogaster nigriceps" = "#009E73")) +
  scale_fill_manual(values = c("Crematogaster sjostedti" = "#E69F00",
                               "Crematogaster mimosae"   = "#56B4E9",
                               "Crematogaster nigriceps" = "#009E73")) +
  # axis labels
  labs(x = "Tree height (cm)",
       y = "Probability of nest occurrence") +
  # clean theme, no gridlines, no legend
  theme_classic() +
  theme(legend.position = "none",
        strip.text = element_text(face = "italic"),
        strip.background = element_blank())
```



2i. Write Caption for figure

Figure 1. Predicted probability of bird nest occurrence as a function of tree height for three acacia ant species in a Kenyan savanna. Lines show model predicted probability of nest occurrence across a gradient of tree heights (cm), with shaded ribbons representing 95% confidence intervals. Points show observed presence (1) or absence (0) of bird nests, jittered slightly for visibility. Each panel represents a different acacia ant species occupying the tree (Crematogaster sjostedti, Crematogaster mimosae, Crematogaster nigriceps). Data from: Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. https://doi.org/10.5281/zenodo.8373322

2j. Calculate model predictions

```
# calculate predicted probability of nest occurrence at 600 cm for each ant species
predict_600 <- ggpredict(model2,
                         terms = c("height_cm [600]",
```

```
                                     "ant_species_full"))

# display output
predict_600
```

```
# Predicted probabilities of case_control

ant_species_full: Crematogaster sjostedti

height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.20 | 0.00, 0.99

ant_species_full: Crematogaster mimosae

height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.74 | 0.48, 0.90

ant_species_full: Crematogaster nigriceps

height_cm | Predicted |     95% CI
----------------------------------
      600 |      1.00 | 1.00, 1.00
```

2k. Interpret your results

As shown in Figure 1, the probability of bird nest occurrence increased with tree height for trees occupied by Crematogaster mimosae and Crematogaster nigriceps, while trees occupied by Crematogaster sjostedti showed little increase in nest probability with height. At 600 cm, the tallest tree in the dataset, the predicted probability of nest occurrence was 0.20 for C. sjostedti, 0.74 for C. mimosae, and 1.00 for C. nigriceps trees (Figure 1). The substantially higher probability of nest occurrence in trees occupied by C. mimosae and C. nigriceps is likely explained by the fact that these species aggressively defend their host trees against mammalian herbivores and predators, which may also protect bird nests from disturbance and predation (Lujan et al. 2023). Additionally, C. nigriceps prunes apical buds to create denser canopies, which likely provides birds with additional concealment from nest predators such as snakes and raptors (Lujan et al. 2023). In contrast, C. sjostedti provides only minimal protection against herbivory, making those trees far less attractive nest sites regardless of tree height (Lujan et al. 2023).

Problem 3. Affective and exploratory visualizations

3a. Comparing visualizations

How are the visualizations different? The exploratory visualizations from Homework 2 used standard statistical plot types, a jitter plot with means and error bars comparing focus level between work and non-work days and a scatterplot of water intake versus focus level, while the affective visualization from Homework 3 represented each day as a hand-drawn drinking glass where fill level encoded water intake and water color (green to blue) encoded focus level, inspired by the Dear Data project. The exploratory plots prioritize precise visual comparison of values across groups, while the affective visualization prioritizes an embodied, personal feeling of each day's energy and hydration.

What similarities do you see? All visualizations represent the same two core variables — daily water intake and focus level — and all show individual observations rather than aggregated summaries alone. Both the exploratory and affective visualizations also incorporate work day as an additional variable, encoded through color grouping in the exploratory plots and through sun or moon symbols in the affective visualization.

What patterns do you see? In the Homework 2 exploratory plots, focus level appeared slightly lower on work days compared to non-work days, and higher water intake appeared associated with higher focus level. The effective visualization from Homework 3 showed the same general pattern — glasses with more water fill tended to have cooler, bluer colors indicating higher focus. The patterns are consistent across visualizations because all draw from the same underlying data set, though the effective visualization makes the day-to-day variation feel more personal and tangible than the summary statistics in the exploratory plots.

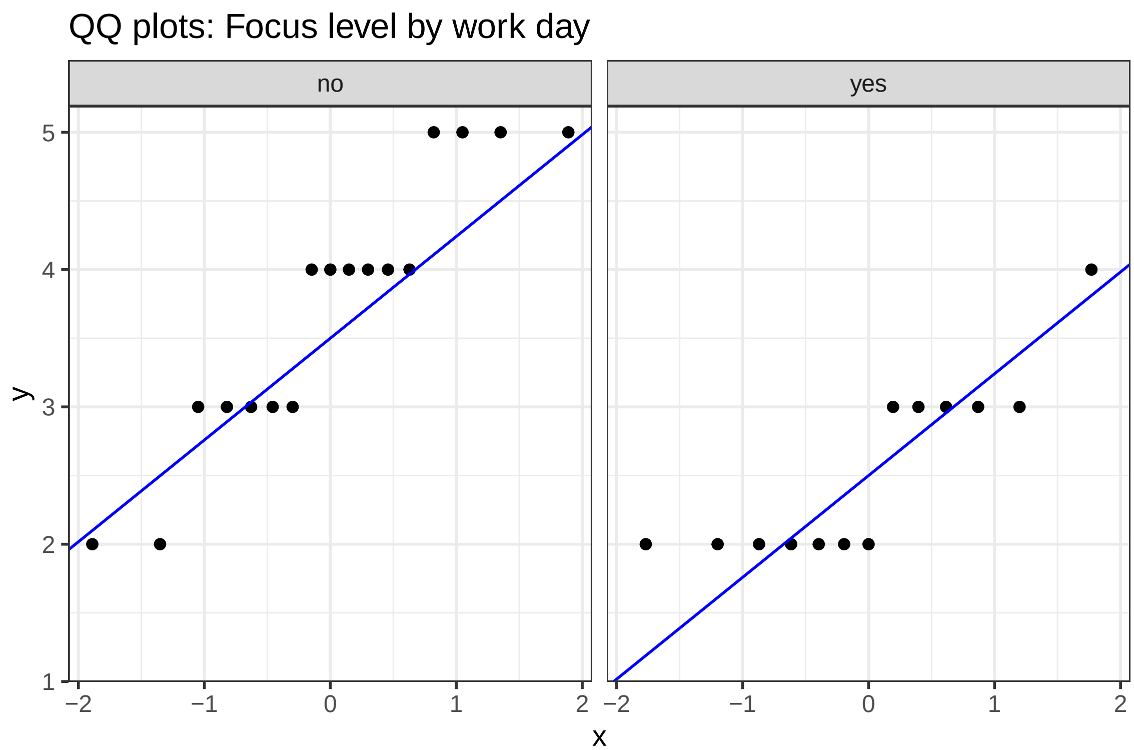
What feedback did you get in week 9? In week 9 workshop, feedback suggested making the legend clearer so viewers could decode what the fill level and water color represented without prior context. I tried to implement this by including a hand-drawn legend on the back of the postcard indicating what fill levels and colors corresponded to specific water intake and focus values, though the legend remains somewhat informal given the hand-drawn medium of the piece.

3b. Designing an analysis

I propose a Welch's two-sample t-test comparing mean focus level between work days and non-work days. This analysis is appropriate because my research question asks whether focus level differs depending on whether the day is a work day; my response variable (focus level, 1–5 scale) is continuous, and my predictor variable (work day: yes/no) is categorical with two levels. A t-test is the appropriate method for comparing means between two independent groups when the response variable is continuous and approximately normally distributed, which I will verify with QQ plots before running the test.

3c. Checking assumptions

```
# check normality of focus level for each work day group using QQ plots
ggplot(data = my_data, # use personal tracking data
       aes(sample = focus_level)) + # specify focus level as the sample variable
  geom_qq() + # add QQ plot points
  geom_qq_line(color = "blue") + # add reference line in blue to assess normality
  facet_wrap(~work_day) + # create separate panels for work and non-work days
  labs(title = "QQ plots: Focus level by work day") + # add descriptive title
  theme_bw() # apply clean black and white theme
```



Running the analysis

```
# run Welch's two-sample t-test comparing focus level between work and non-work days
t_test_result <- t.test(focus_level ~ work_day, # focus level as response, work day as predi
                        data = my_data, # use personal tracking data
                        var.equal = FALSE) # specify Welch's t-test, does not assume equal va
```

```r
# display result
t_test_result
```

```

	Welch Two Sample t-test

data:  focus_level by work_day
t = 3.878, df = 27.613, p-value = 0.0005929
alternative hypothesis: true difference in means between group no and group yes is not equal
95 percent confidence interval:
 0.5503826 1.7844591
sample estimates:
 mean in group no mean in group yes 
         3.705882          2.538462 
```

Effect size

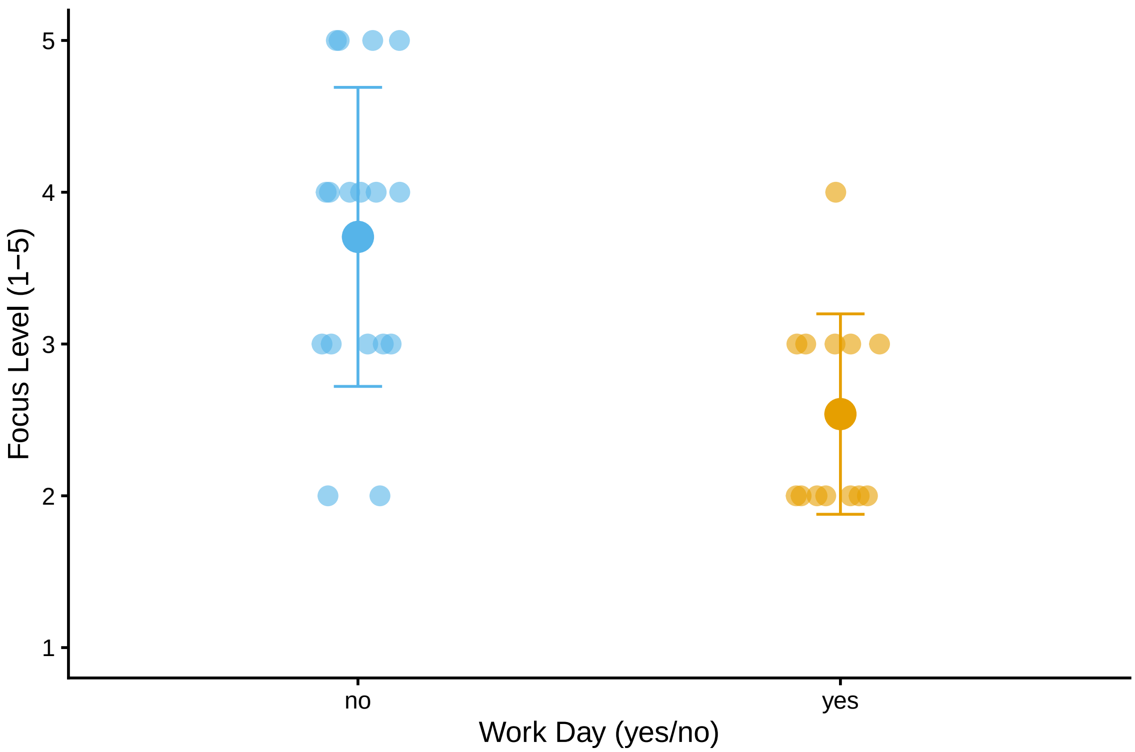
```r
# calculate Cohen's d effect size to quantify magnitude of difference
cohens_d(focus_level ~ work_day, # focus level as response, work day as predictor
         data = my_data) # use personal tracking data
```

```
# A tibble: 1 x 7
  .y.         group1 group2 effsize    n1    n2 magnitude
* <chr>       <chr>  <chr>    <dbl> <int> <int> <ord>    
1 focus_level no     yes       1.39    17    13 large    
```

3d. Visualization

```r
# summarize mean and SD by work day for plotting
my_data_summary <- my_data |>
  group_by(work_day) |> # group observations by work day status
  summarize(mean_focus = mean(focus_level), # calculate mean focus level per group
            sd_focus = sd(focus_level)) # calculate standard deviation per group

# plot focus level by work day with raw data, means, and SD error bars
ggplot() +
  # add individual data points jittered horizontally to reduce overlap
```

```
geom_jitter(data = my_data,
            aes(x = work_day,
                y = focus_level,
                color = work_day),
            width = 0.1, height = 0, # jitter only horizontally, not vertically
            size = 3, alpha = 0.6) + # set point size and transparency
# add larger point showing group mean
geom_point(data = my_data_summary,
           aes(x = work_day,
               y = mean_focus,
               color = work_day),
           size = 5) + # larger size to distinguish mean from raw data points
# add error bars showing plus or minus 1 standard deviation
geom_errorbar(data = my_data_summary,
              aes(x = work_day,
                  ymin = mean_focus - sd_focus, # lower bound of error bar
                  ymax = mean_focus + sd_focus, # upper bound of error bar
                  color = work_day),
              width = 0.1) + # set width of error bar caps
# apply colorblind-friendly colors: blue for non-work, orange for work days
scale_color_manual(values = c("no" = "#56B4E9",
                              "yes" = "#E69F00")) +
# set y-axis to show full 1-5 scale with breaks at each integer
scale_y_continuous(limits = c(1, 5),
                   breaks = c(1, 2, 3, 4, 5)) +
# add descriptive axis labels
labs(x = "Work Day (yes/no)",
     y = "Focus Level (1-5)") +
theme_classic() + # remove gridlines and grey background
theme(legend.position = "none") # remove legend since colors are self-explanatory
```



3e. Write a caption

Figure 2. Daily focus level (1-5 scale) on work days versus non-work days across 30 days of personal observation. Small points show individual daily focus level observations, jittered horizontally to reduce overlap. Large points show the group mean and error bars represent plus or minus 1 standard deviation. Blue points represent non-work days and orange points represent work days.

3f. Write about your results

Non-work days had a higher mean focus level (mean = 3.71, n = 17) compared to work days (mean = 2.54, n = 13), a difference of about 1.17 points on the 1-5 scale. This difference was statistically significant and large in magnitude (Welch's t-test: $t(27.61) = 3.88$, $p < 0.001$, Cohen's $d = 1.39$), suggesting that whether or not a day is a work day has a strong association with how focused I feel. This pattern may reflect the increased demands and reduced autonomy over daily schedule that come with work days, which could make it harder to sustain attention and mental clarity throughout the day.